

CSCE 775: Deep Reinforcement Learning

Homework #1 — Solutions

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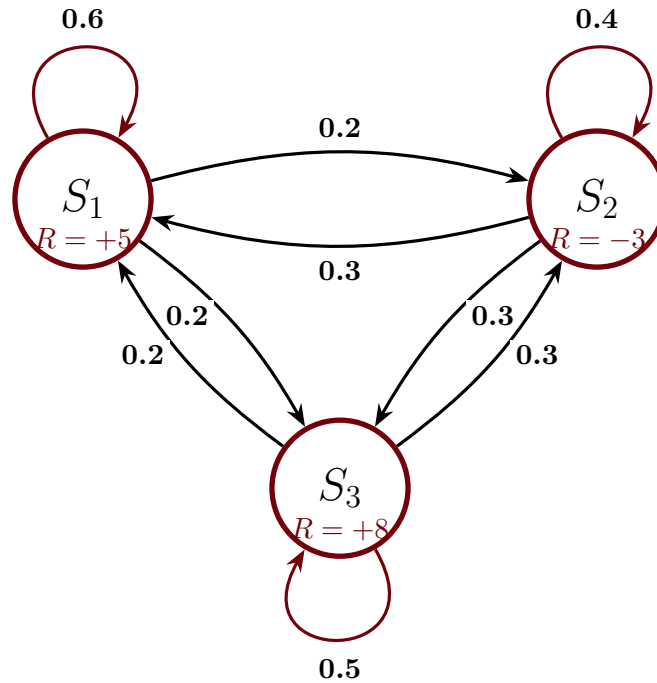
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Problem 1: MRP State Values

Consider the following Markov Reward Process with three states $\mathcal{S} = \{S_1, S_2, S_3\}$.



The transition matrix and reward vector are:

$$\mathcal{P} = \begin{pmatrix} 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} +5 \\ -3 \\ +8 \end{pmatrix}$$

- (a) Write the Bellman equation for this MRP in matrix form for a general discount factor γ . That is, express $\mathbf{V}$ as a function of $\mathbf{R}$, γ , and $\mathcal{P}$.

Solution 1(a)

The Bellman equation for an MRP is $\mathbf{V} = \mathbf{R} + \gamma\mathcal{P}\mathbf{V}$. Rearranging:

$$(\mathbf{I} - \gamma\mathcal{P})\mathbf{V} = \mathbf{R}$$

Results

$$\mathbf{V} = (\mathbf{I} - \gamma\mathcal{P})^{-1}\mathbf{R}$$

- (b) Form the matrix $(\mathbf{I} - \gamma\mathcal{P})$ explicitly for $\gamma = 0.8$. Show all entries.

Solution 1(b)

$$\mathbf{I} - 0.8\mathcal{P} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - 0.8 \begin{pmatrix} 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}$$

Results

$$\mathbf{I} - 0.8\mathcal{P} = \begin{pmatrix} 0.52 & -0.16 & -0.16 \\ -0.24 & 0.68 & -0.24 \\ -0.16 & -0.24 & 0.60 \end{pmatrix}$$

- (c) Solve the 3×3 linear system $(\mathbf{I} - \gamma\mathcal{P})\mathbf{V} = \mathbf{R}$ for $\gamma = 0.8$ to find $V(S_1)$, $V(S_2)$, and $V(S_3)$. Show your work.

Solution 1(c) — Setup

We solve via Cramer's rule:

$$\begin{pmatrix} 0.52 & -0.16 & -0.16 \\ -0.24 & 0.68 & -0.24 \\ -0.16 & -0.24 & 0.60 \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -3 \\ 8 \end{pmatrix}$$

Step 1: Determinant of A

$$\begin{aligned} \det(A) &= 0.52(0.68 \cdot 0.60 - (-0.24)(-0.24)) \\ &\quad - (-0.16)((-0.24)(0.60) - (-0.24)(-0.16)) \\ &\quad + (-0.16)((-0.24)(-0.24) - (0.68)(-0.16)) \\ &= 0.52(0.4080 - 0.0576) + 0.16(-0.1440 - 0.0384) - 0.16(0.0576 + 0.1088) \\ &= 0.52(0.3504) + 0.16(-0.1824) - 0.16(0.1664) \\ &= 0.18221 - 0.02918 - 0.02662 = \mathbf{0.12640} \end{aligned}$$

Step 2: Numerator determinants

For V_1 (replace column 1 with $\mathbf{R}$):

$$\det(A_1) = 5(0.3504) + 0.16(0.12) - 0.16(-4.72) = 1.7520 + 0.0192 + 0.7552 = \mathbf{2.5264}$$

For V_2 (replace column 2 with $\mathbf{R}$):

$$\det(A_2) = 0.52(0.12) - 5(-0.1824) - 0.16(-2.4) = 0.0624 + 0.9120 + 0.3840 = \mathbf{1.3584}$$

For V_3 (replace column 3 with $\mathbf{R}$):

$$\det(A_3) = 0.52(4.72) + 0.16(-2.4) + 5(0.1664) = 2.4544 - 0.3840 + 0.8320 = \mathbf{2.9024}$$

Results

$$V(S_1) = \frac{2.5264}{0.1264} \approx 19.99,$$

$$V(S_2) = \frac{1.3584}{0.1264} \approx 10.75,$$

$$V(S_3) = \frac{2.9024}{0.1264} \approx 22.96$$

Problem 2: MRP Discount Factor Analysis

Use the same MRP from Problem 1.

- (a) Compute $V(S_1)$, $V(S_2)$, and $V(S_3)$ for $\gamma = 0$. Briefly explain why the answer is immediate.

Solution 2(a)

When $\gamma = 0$, the Bellman equation becomes $\mathbf{V} = \mathbf{R} + 0 \cdot \mathcal{P}\mathbf{V} = \mathbf{R}$. The agent is completely myopic—it only values the immediate reward with no consideration of future states.

Results

$$V(S_1) = 5, \quad V(S_2) = -3, \quad V(S_3) = 8$$

- (b) Compute $V(S_1)$, $V(S_2)$, and $V(S_3)$ for $\gamma = 0.5$.

Solution 2(b)

Form $\mathbf{I} - 0.5\mathcal{P}$:

$$\mathbf{I} - 0.5\mathcal{P} = \begin{pmatrix} 0.70 & -0.10 & -0.10 \\ -0.15 & 0.80 & -0.15 \\ -0.10 & -0.15 & 0.75 \end{pmatrix}$$

Using Cramer's rule (same procedure as Problem 1(c)):

$$\det(\mathbf{I} - 0.5\mathcal{P}) = 0.38125$$

$$\det(A_1) = 3.3775, \quad \det(A_2) = 0.0525, \quad \det(A_3) = 4.5275$$

Results

$$V(S_1) \approx 8.86, \quad V(S_2) \approx 0.14, \quad V(S_3) \approx 11.88$$

- (c) For approximately what value of γ does $V(S_2)$ cross from negative to positive? Set $V(S_2) = 0$ in the Bellman equation and solve.

Solution 2(c) — Setting up the constraint

Set $V(S_2) = 0$ in the Bellman equation. From the second row:

$$V(S_2) = -3 + \gamma[0.3V(S_1) + 0.4 \cdot 0 + 0.3V(S_3)] = 0 \implies V(S_1) + V(S_3) = \frac{10}{\gamma}$$

From the first and third rows (with $V(S_2) = 0$):

$$V(S_1)(1 - 0.6\gamma) = 5 + 0.2\gamma V(S_3), \quad V(S_3)(1 - 0.5\gamma) = 8 + 0.2\gamma V(S_1)$$

Solving the quadratic

Substituting $V(S_1) = 10/\gamma - V(S_3)$ into the third equation yields $V(S_3) = 10/(1 - 0.3\gamma)$. Substituting both into the first equation and simplifying:

$$7.3\gamma^2 - 24\gamma + 10 = 0$$

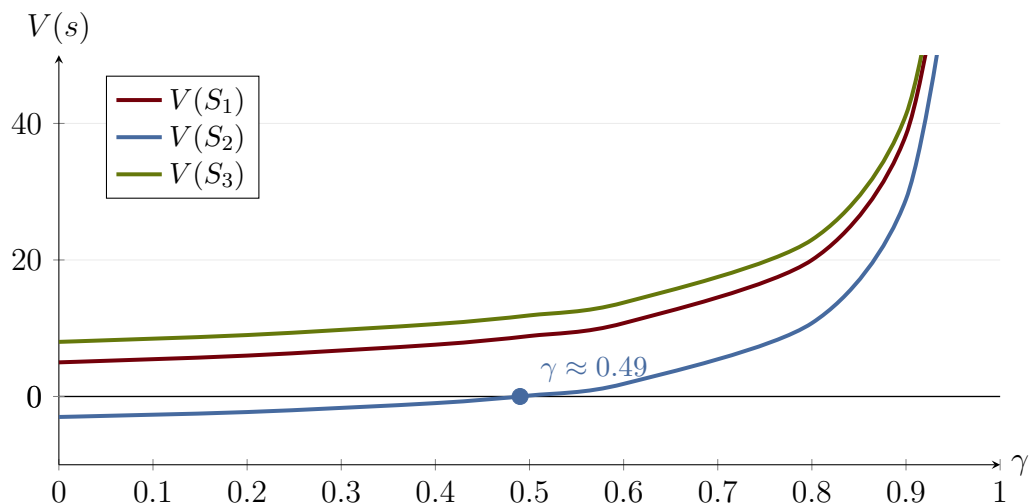
Applying the quadratic formula:

$$\gamma = \frac{24 - \sqrt{576 - 292}}{14.6} = \frac{24 - \sqrt{284}}{14.6} = \frac{24 - 16.852}{14.6}$$

Results

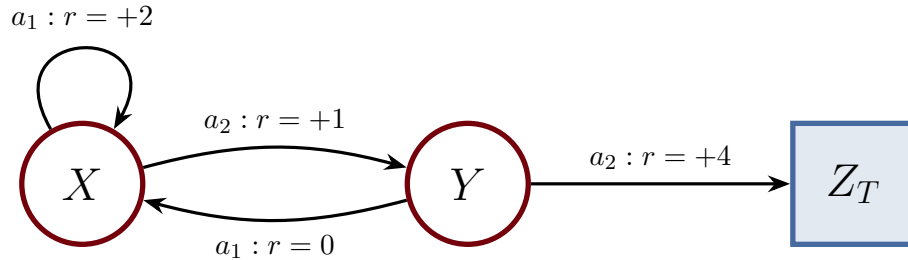
$$\boxed{\gamma \approx 0.49}$$

- (d) On the axes below, sketch the curves $V(S_1)$, $V(S_2)$, and $V(S_3)$ as functions of $\gamma \in [0, 1]$. Label each curve and mark the zero-crossing from part (c).



Problem 3: Value Iteration by Hand

Consider the following MDP with states $\{X, Y\}$ and a terminal state Z_T , discount factor $\gamma = 0.9$. Each non-terminal state has two actions: a_1 and a_2 . All transitions are deterministic.



Transition and reward summary:

State	Action	Next State	Reward
X	a_1	X	$+2$
X	a_2	Y	$+1$
Y	a_1	X	0
Y	a_2	Z_T	$+4$

- (a) Initialize $V_0(X) = V_0(Y) = 0$. Perform **3 iterations** of value iteration. Fill in the table below. Show computation for each cell.

Solution 3(a) — Iteration 1

Update rule: $V_{k+1}(s) = \max_a [R(s, a) + \gamma V_k(s')]$. Starting from $V_0(X) = V_0(Y) = 0$:

$$Q(X, a_1) = 2 + 0.9 \cdot V_0(X) = 2 + 0 = 2$$

$$Q(X, a_2) = 1 + 0.9 \cdot V_0(Y) = 1 + 0 = 1$$

$$V_1(X) = \max(2, 1) = 2, \quad \pi_1(X) = a_1$$

$$Q(Y, a_1) = 0 + 0.9 \cdot V_0(X) = 0$$

$$Q(Y, a_2) = 4 + 0.9 \cdot 0 = 4$$

$$V_1(Y) = \max(0, 4) = 4, \quad \pi_1(Y) = a_2$$

Iteration 2

Using $V_1(X) = 2$, $V_1(Y) = 4$:

$$Q(X, a_1) = 2 + 0.9(2) = 3.8$$

$$Q(X, a_2) = 1 + 0.9(4) = 4.6$$

$$V_2(X) = 4.6, \quad \pi_2(X) = a_2$$

$$Q(Y, a_1) = 0 + 0.9(2) = 1.8$$

$$Q(Y, a_2) = 4 + 0.9(0) = 4$$

$$V_2(Y) = 4, \quad \pi_2(Y) = a_2$$

Iteration 3

Using $V_2(X) = 4.6$, $V_2(Y) = 4$:

$$Q(X, a_1) = 2 + 0.9(4.6) = 6.14$$

$$Q(X, a_2) = 1 + 0.9(4) = 4.6$$

$$V_3(X) = 6.14, \quad \pi_3(X) = a_1$$

$$Q(Y, a_1) = 0 + 0.9(4.6) = 4.14$$

$$Q(Y, a_2) = 4 + 0 = 4$$

$$V_3(Y) = 4.14, \quad \pi_3(Y) = a_1$$

Results

Iter k	$V_k(X)$	$\pi_k(X)$	$V_k(Y)$	$\pi_k(Y)$
0	0	—	0	—
1	2.00	a_1	4.00	a_2
2	4.60	a_2	4.00	a_2
3	6.14	a_1	4.14	a_1

- (b) After convergence, what is the optimal policy π^* ? State $\pi^*(X)$ and $\pi^*(Y)$ and justify briefly.

Solution 3(b)

At convergence, the optimal policy is $\pi^*(X) = a_1$ and $\pi^*(Y) = a_1$. From state X , taking a_1 yields reward +2 and stays in X , accumulating $2/(1 - 0.9) = 20$ in total discounted reward. Taking a_2 would move to Y with reward +1, which is suboptimal. From state Y , taking a_1 transitions to X (reward 0) and then the agent collects the stream of +2's; this is worth $0.9 \cdot 20 = 18$, which exceeds a_2 's value of 4.

Results

$$\pi^*(X) = a_1, \quad \pi^*(Y) = a_1$$

- (c) Compute the exact optimal values $V^*(X)$ and $V^*(Y)$ by solving the Bellman optimality equations simultaneously.

Solution 3(c)

Assuming $\pi^*(X) = a_1$ and $\pi^*(Y) = a_1$ (verified by checking all Q-values):

$$V^*(X) = 2 + 0.9 V^*(X) \implies V^*(X)(1 - 0.9) = 2 \implies V^*(X) = 20$$

$$V^*(Y) = 0 + 0.9 V^*(X) = 0.9 \cdot 20 = 18$$

Verification: $Q(X, a_2) = 1 + 0.9(18) = 17.2 < 20 = Q(X, a_1) \checkmark$
 $Q(Y, a_2) = 4 + 0 = 4 < 18 = Q(Y, a_1) \checkmark$

Results

$$V^*(X) = 20, \quad V^*(Y) = 18$$

Problem 4: Grid World Policy

Consider the 3×3 grid MDP below. The agent can move **Up**, **Down**, **Left**, **Right**. Moves that would leave the grid keep the agent in place. All transitions are deterministic. Discount factor $\gamma = 1$.

(0, 2)	(1, 2)	Goal $r = +10$
(0, 1)	Wall	Trap $r = -5$
Start	(1, 0)	(2, 0)

Rules:

- Every move costs $r = -1$ (step penalty), except moves into Goal or Trap.
- Reaching the **Goal** (2, 2): reward $r = +10$, episode ends.
- Reaching the **Trap** (2, 1): reward $r = -5$, episode ends.
- The **Wall** (1, 1) is impassable; moves into it keep the agent in place (still costs -1).
- Goal and Trap are terminal states.

- (a) List the reward $R(s, a, s')$ for every non-terminal state s and every action a . Organize your answer clearly.

Solution 4(a)				
State	Up	Down	Left	Right
(0, 0)	(0, 1), $r=-1$	(0, 0), $r=-1$	(0, 0), $r=-1$	(1, 0), $r=-1$
(1, 0)	Wall $\rightarrow$ (1, 0), $r=-1$	(1, 0), $r=-1$	(0, 0), $r=-1$	(2, 0), $r=-1$
(2, 0)	Trap, $r=-5$	(2, 0), $r=-1$	(1, 0), $r=-1$	(2, 0), $r=-1$
(0, 1)	(0, 2), $r=-1$	(0, 0), $r=-1$	(0, 1), $r=-1$	Wall $\rightarrow$ (0, 1), $r=-1$
(0, 2)	(0, 2), $r=-1$	(0, 1), $r=-1$	(0, 2), $r=-1$	(1, 2), $r=-1$
(1, 2)	(1, 2), $r=-1$	Wall $\rightarrow$ (1, 2), $r=-1$	(0, 2), $r=-1$	Goal, $r=+10$

- (b) Starting from $V_0(s) = 0$ for all states, perform **2 iterations** of value iteration.

Solution 4(b) — Iteration 1

All $V_0 = 0$, so $V_1(s) = \max_a R(s, a, s')$:

- (0,0): all actions give -1 . $V_1 = -1$.
- (1,0): all actions give -1 . $V_1 = -1$.
- (2,0): Up= -5 , others= -1 . $V_1 = -1$.
- (0,1): all actions give -1 . $V_1 = -1$.
- (0,2): all actions give -1 . $V_1 = -1$.
- (1,2): Right= $+10$, others= -1 . $V_1 = 10$.

Iteration 2

Using V_1 values above, with $\gamma = 1$:

- (0,0): Up: $-1 + V_1(0,1) = -2$; Down: $-1 + V_1(0,0) = -2$; Left: -2 ; Right: $-1 + V_1(1,0) = -2$. $V_2 = -2$.
- (1,0): Up(wall): $-1 + (-1) = -2$; Down: -2 ; Left: -2 ; Right: $-1 + (-1) = -2$. $V_2 = -2$.
- (2,0): Up=Trap: $-5 + 0 = -5$; Down: -2 ; Left: -2 ; Right: -2 . $V_2 = -2$.
- (0,1): Up: $-1 + (-1) = -2$; Down: -2 ; Left: -2 ; Right(wall): -2 . $V_2 = -2$.
- (0,2): Right: $-1 + V_1(1,2) = -1 + 10 = 9$; others: -2 . $V_2 = 9$.
- (1,2): Right=Goal: $10 + 0 = 10$; Up: $-1 + 10 = 9$; Down(wall): $-1 + 10 = 9$; Left: $-1 + (-1) = -2$. $V_2 = 10$.

After Iteration 1:

-1	10	G
-1	W	T
-1	-1	-1

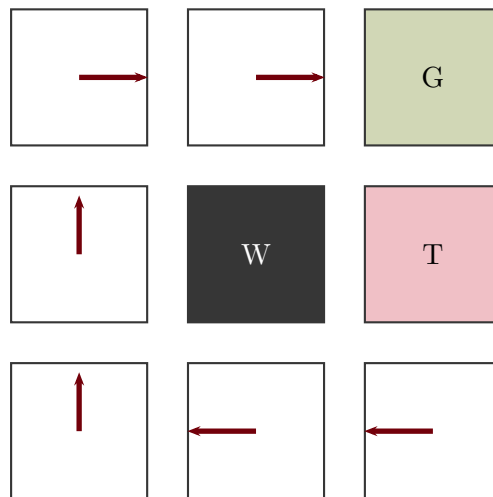
After Iteration 2:

9	10	G
-2	W	T
-2	-2	-2

- (c) On the grid below, draw the optimal policy (one arrow per non-terminal cell indicating the best action).

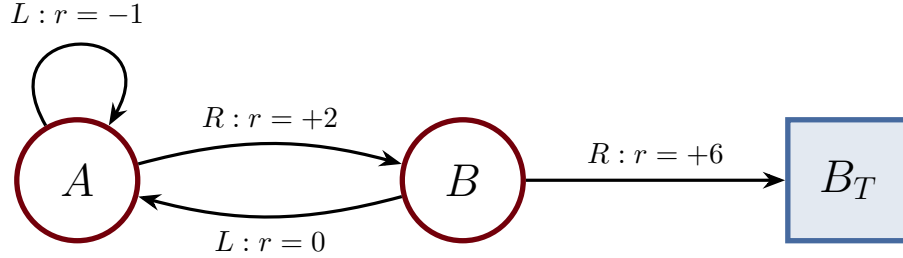
Solution 4(c)

The optimal path avoids the Trap and routes through the left column up to (0, 2), then right along the top row to Goal. The optimal policy is:



Problem 5: Q-Learning Episode Trace

Consider the following 2-state MDP with states $\{A, B\}$ and terminal state B_T . Each state has two actions: L and R .



All transitions are deterministic. Use **Q-learning** with $\alpha = 0.2$, $\gamma = 1$, and $\varepsilon = 0$ (pure greedy; break ties alphabetically by action name, i.e., L before R).

Initialize all Q-values to zero: $Q(A, L) = Q(A, R) = Q(B, L) = Q(B, R) = 0$.

The three episode trajectories are given (not chosen by your policy—just apply the updates):

Episode 1: $A \xrightarrow{R} B \xrightarrow{R} B_T$

Episode 2: $A \xrightarrow{R} B \xrightarrow{R} B_T$

Episode 3: $A \xrightarrow{L} A \xrightarrow{R} B \xrightarrow{R} B_T$

- (a) For each step of each episode, compute the Q-learning update. Fill in the table.

Recall: $Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$

Solution 5(a) — Episode 1

Step 1: $s = A, a = R, s' = B, r = +2$

$$Q(A, R) \leftarrow 0 + 0.2[2 + 1 \cdot \max(0, 0) - 0] = 0.2(2) = 0.40$$

Step 2: $s = B, a = R, s' = B_T, r = +6$

$$Q(B, R) \leftarrow 0 + 0.2[6 + 0 - 0] = 1.20$$

Episode 2

Step 1: $s = A, a = R, s' = B, r = +2$. Now $\max_{a'} Q(B, a') = 1.20$.

$$Q(A, R) \leftarrow 0.40 + 0.2[2 + 1.20 - 0.40] = 0.40 + 0.2(2.80) = 0.96$$

Step 2: $s = B, a = R, s' = B_T, r = +6$

$$Q(B, R) \leftarrow 1.20 + 0.2[6 + 0 - 1.20] = 1.20 + 0.2(4.80) = 2.16$$

Episode 3

Step 1: $s = A, a = L, s' = A, r = -1$. $\max_{a'} Q(A, a') = 0.96$.

$$Q(A, L) \leftarrow 0 + 0.2[-1 + 0.96 - 0] = 0.2(-0.04) = -0.008$$

Step 2: $s = A, a = R, s' = B, r = +2$. $\max_{a'} Q(B, a') = 2.16$.

$$Q(A, R) \leftarrow 0.96 + 0.2[2 + 2.16 - 0.96] = 0.96 + 0.2(3.20) = 1.60$$

Step 3: $s = B, a = R, s' = B_T, r = +6$

$$Q(B, R) \leftarrow 2.16 + 0.2[6 + 0 - 2.16] = 2.16 + 0.2(3.84) = 2.928$$

Results

Ep.	Step	s	a	s'	r	New $Q(s, a)$
1	1	A	R	B	+2	$Q(A, R) = 0.400$
1	2	B	R	B_T	+6	$Q(B, R) = 1.200$
2	1	A	R	B	+2	$Q(A, R) = 0.960$
2	2	B	R	B_T	+6	$Q(B, R) = 2.160$
3	1	A	L	A	-1	$Q(A, L) = -0.008$
3	2	A	R	B	+2	$Q(A, R) = 1.600$
3	3	B	R	B_T	+6	$Q(B, R) = 2.928$

(b) State all four Q-values after the 3 episodes are complete.

Results

$$Q(A, L) = -0.008, \quad Q(A, R) = 1.600, \quad Q(B, L) = 0.000, \quad Q(B, R) = 2.928$$

(c) What is the greedy policy $\pi(s) = \arg \max_a Q(s, a)$ for each state?

Results

$$\pi(A) = R \quad (\text{since } 1.600 > -0.008), \quad \pi(B) = R \quad (\text{since } 2.928 > 0)$$

Problem 6: Exploration vs. Exploitation

- (a) Suppose an agent is in a state with three available actions $\{a_1, a_2, a_3\}$ and current Q-values:

$$Q(s, a_1) = 5.0, \quad Q(s, a_2) = 3.2, \quad Q(s, a_3) = 7.1$$

Under an ε -greedy policy with $\varepsilon = 0.4$, compute the probability of selecting each action.

Solution 6(a)

Under ε -greedy with $|\mathcal{A}| = 3$ actions:

- The greedy action is a_3 (highest Q-value of 7.1).
- Each action gets a base probability of $\varepsilon/|\mathcal{A}| = 0.4/3 = 2/15$ from random exploration.
- The greedy action gets an additional $1 - \varepsilon = 0.6$ probability.

Results

$$P(a_1) = \frac{2}{15} \approx 0.133, \quad P(a_2) = \frac{2}{15} \approx 0.133, \quad P(a_3) = 0.6 + \frac{2}{15} = \frac{11}{15} \approx 0.733$$

Verification: $\frac{2}{15} + \frac{2}{15} + \frac{11}{15} = \frac{15}{15} = 1 \checkmark$

- (b) Suppose ε starts at $\varepsilon_0 = 1.0$ and decays multiplicatively each episode by a factor of 0.99:

$$\varepsilon_k = \varepsilon_0 \cdot 0.99^k$$

After how many episodes k is $\varepsilon_k < 0.05$? Solve exactly using logarithms.

Solution 6(b)

$$\begin{aligned} 0.99^k &< 0.05 \\ k \ln(0.99) &< \ln(0.05) \\ k &> \frac{\ln(0.05)}{\ln(0.99)} \quad (\text{inequality flips since } \ln(0.99) < 0) \\ k &> \frac{-2.9957}{-0.01005} = 298.07 \end{aligned}$$

Results

$$k = 299$$

is the first episode where $\varepsilon_k < 0.05$. ($\varepsilon_{299} = 0.99^{299} \approx 0.0495 < 0.05$; $\varepsilon_{298} \approx 0.0500 \geq 0.05$.)

- (c) In 3–5 sentences, explain: Why is exploration important in the early stages of learning? Why is it beneficial to decay ϵ over time rather than keeping it fixed?

Solution 6(c)

In the early stages of learning, the agent's Q-value estimates are inaccurate (typically initialized to zero or random values), so the greedy action with respect to current estimates is unlikely to be truly optimal. Exploration allows the agent to visit state–action pairs it has not yet tried, gathering information that improves Q-value estimates across the entire state–action space. Without sufficient exploration, the agent may converge to a suboptimal policy because it never discovers higher-reward paths.

Decaying ϵ over time is beneficial because it allows the agent to transition from an *information-gathering* phase to an *exploitation* phase. As training progresses and Q-values become more accurate, continued random exploration wastes reward by choosing known-suboptimal actions. A decaying schedule balances the exploration–exploitation tradeoff: explore broadly when uncertain, then increasingly commit to the learned policy as confidence grows.